Analytical Cume
October 22, 1999

Mifflin and associates described a membrane electrode for the quantitative analysis of penicillin in which the enzyme penicillinase is immobilized in a polyacrylamide gel that is coated on a glass pH electrode. The following date were collected for a series of penicillin standards.

[Penicillin] (M)	Potential (mV)
1.0×10^{-2}	220
2.0×10^{-3}	204
1.0×10^{-3}	190
2.0×10^{-4}	153
1.0×10^{-4}	135
1.0×10^{-5}	96
1.0×10^{-6}	80

1. (10pts) Construct a calibration curve for the electrode which should theoretically produce a straight line.
2. (10pts) Report the range of concentrations in which a linear response is observed
3. (15 pts) Give the equation for the calibration curve in the linear range.
4. (15pts) Determine the concentration of penicillin in a sample that yields a potential of 142 mV.
5. (15pts) Give a description of how penicillinase must work to produce a different voltage for each concentration of penicillin.

For the analysis of toothpaste a 0.3619 g sample was transferred to a 100mL volumetric flask along with 50 mL of buffer and diluted to volume with distilled water. Three 20.0 mL aliquots were removed, and the potential was measure with an F^- ISE using an SCE as a reference electrode. Five Separate 1.00 mL addition of 100.0 ppm solution of F^- were added to each, measuring the potential following each addition. The collected data is as follows:

mL of 100.0 ppm F^- Standard added	Potential (mV) Trial 1	Trial 2	Trial 3
0.00	-55	-54	-55
1.00	-82	-82	-83
2.00	-94	-94	-94
3.00	-102	-103	-102
4.00	-108	-108	-109
5.00	-112	-112	-113

6.(35pts) Determine the ppm of F^- in the tooth paste sample.